



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Nmap Fundamentals & Authorized Network Discovery

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1. THEORY

1.1 Answer all 20 questions in your own words

1. What does Nmap stand for?

Nmap stands for Network Mapper.

2. What is Nmap used for?

Nmap is used to discover hosts, check ports, and identify services on authorized systems.

3. What is network discovery?

Network discovery means finding and identifying devices or hosts on a network.

4. What is a port scan?

A port scan checks which network ports are open, closed, or filtered on a target.

5. What does an open port generally indicate?

A port open generally means a service is accepting connections on that port.

6. Does an open port automatically mean vulnerability? Explain.

No. An open port does not automatically mean the system is vulnerable. Risk depends on the service, version, configuration, security and exposure.

7. What does closed mean conceptually?

Closed means the host can be reached, but no service is accepting connections on that port.

8. What does filtered mean conceptually?

Filtered means Nmap cannot clearly determine the port's state because a firewall or security control may be blocking the response.

9. What does `nmap 127.0.0.1` do?

It scans local computer (localhost) to check its available ports.

10. What does `-p` do?

`-p` is used to select specific ports or port ranges to scan.

11. What does `-sV` do?

`-sV` attempts to identify the service and its version running on open ports.

12. What does `-oN` do?

`-oN` saves the Nmap results in a normal readable text file.

13. Why should scan results be documented?

Results should be documented to keep evidence, review findings, and maintain a record of the assessment.

14. What is cybersecurity scope?

Scope defines which systems or IP addresses are allowed to be assessed and which are not.

15. Why must authorization exist before scanning?

Authorization is required because scanning systems without permission can be unauthorized and potentially disruptive.

16. What is the difference between scanning and exploitation?

Scanning collects information about a system. Exploitation attempts to use a weakness to gain access or affect the system.

17. Why is localhost a good beginner target?

Localhost is good target because it is your own computer and is safe for learning basic scanning.

18. How do ports relate to services?

Ports act as communication endpoints for services. For example, HTTP commonly use port 80 and HTTPS commonly uses 443.

19. Why are unexpected listening services worth reviewing?

They should be reviewed because and unnecessary or unknown service may increase the system's exposure.

20. Explain a responsible Nmap assessment workflow.

A responsible workflow is: get authorized - define scope – identify the target – perform basic discovery – check ports – identify services – document results – review the findings.

2.PRACTICAL LAB

Your First Authorized Nmap Assessment Environment: your Kali Linux VM (Nmap is preinstalled on Kali). Target: localhost OR a second VM you personally control/are authorized to use.

TASK 1 — `nmap --version` → record Nmap Version.

```
(kali㉿kali)-[/home/kali]
PS> nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.2.2 libssh2-1.11.0 libz-1.3.1 libpcap-1.10.4 nmap-libdnet-1.12
              ipv6
Compiled without:
Available nsock engines: epoll poll select
```

TASK 2 — `nmap --help` → what does `-p` control? (Ports.)

```
-b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
-p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
--exclude-ports <port ranges>: Exclude the specified ports from scanning
-F: Fast mode - Scan fewer ports than the default scan
```

TASK 3 — `nmap 127.0.0.1` → record the number of open ports. ZERO open ports is perfectly acceptable — do NOT start unnecessary services to "get results".

```
(kali㉿kali)-[/home/kali]
PS> nmap 127.0.0.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-21 11:14 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00033s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

TASK 4 — `nmap -p 22,80,443 127.0.0.1` → build your table: Port / State / Service.

```
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds

(kali㉿kali)-[/home/kali]
PS> nmap -p 22,80,443 127.0.0.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-21 11:14 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00048s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

TASK 5 — `nmap -sV 127.0.0.1` → record any detected service/version; if none: "No service/version detected".

```
(kali㉿kali)-[/home/kali]
PS> nmap -sV 127.0.0.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-21 11:16 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00018s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

TASK 6 — `nmap -oN day14-scan.txt 127.0.0.1` then `cat day14-scan.txt` → confirm your report was saved.

```
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds

(kali㉿kali)-[/home/kali]
PS> nmap -oN day14-scan.txt 127.0.0.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-21 11:16 EDT
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00019s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds

(kali㉿kali)-[/home/kali]
PS> cat day14-scan.txt
/usr/bin/cat: day14-scan.txt: No such file or directory

(kali㉿kali)-[/home/kali]
PS> cat day14-scan.txt
# Nmap 7.94SVN scan initiated Mon Sep 21 11:16:39 2026 as: /usr/bin/nmap -oN day14-scan.txt 127.0.0.1
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00019s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

# Nmap done at Mon Sep 21 11:16:39 2026 -- 1 IP address (1 host up) scanned in 0.09 seconds
```

TASK 10 — Mini report: TARGET: 127.0.0.1 · AUTHORIZED: YES OPEN PORTS:
_____ SERVICES: _____ OBSERVATIONS: _____ RECOMMENDATION:
_____ (example: disable unnecessary services and restrict exposure per business
need)

Target: 127.0.0.1

Authorized: YES

Open Ports:0

Services: No open services detected

Observations: The localhost system was up and reachable. Nmap scanned 1000 TCP
ports, and all 1000 ports were closed. No service/version was detected.

Recommendation: Review the system regularly and keep unnecessary network services
disabled. If services are enabled in the future, ensure they are required and properly
secured.